Again, let me emphasize that the future measurements do not change anything at all about things that took place in your experiment today; the future measurements do not in any way change the data you collected today. But the future measurements *do* influence the kinds of details you can invoke when you subsequently describe what happened today. Before you have the results of the idler photon measurements, you really can't say anything at all about the which-path history of any given signal photon. However, once you have the results, you conclude that signal photons whose idler partners were successfully used to ascertain which-path information *can* be described as having—years earlier—traveled either left or right. You also conclude that signal photons whose idler partners had their which-path information erased *cannot* be described as having— years earlier—definitely gone one way or the other (a conclusion you can convincingly confirm by using the newly acquired idler photon data to expose the previously hidden interference pattern among this latter class of signal photons). We thus see that the future helps shape the story you tell of the past.

These experiments are a magnificent affront to our conventional notions of space and time. Something that takes place long after and far away from something else nevertheless is vital to our description of that something else. By any classical—commonsense—reckoning, that's, well, crazy. Of course, that's the point: classical reckoning is the wrong kind of reckoning to use in a quantum universe. We have learned from the Einstein-Podolsky-Rosen discussion that quantum physics is not local in space. If you have fully absorbed that lesson—a tough one to accept in its own right—these experiments, which involve a kind of entanglement across space and through time, may not seem thoroughly outlandish. But by the standards of daily experience, they certainly are.

Quantum Mechanics and Experience

For a few days after I first learned about these experiments, I remember feeling elated. I felt I'd been given a glimpse into a veiled side of reality. Common experience—mundane, ordinary, day-to-day activities—suddenly seemed part of a classical charade, hiding the true nature of our quantum world. The world of the everyday suddenly seemed nothing but an inverted magic act, lulling its audience into believing in the usual,